DM IMCQ 1
Spring 2026

Anatomy and Biochemistry

KEY file

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Anatomy 7 Questions

A preterm girl is brought to the physician for failure to pass meconium within 48 hours of birth. The patient has also had progressive abdominal distention and vomiting. Physical exam shows a distended abdomen, and an inability to examine beyond 1cm into the anal canal or flush the distal rectum. No discharge is seen from the vaginal canal. Abdominal X-ray shows dilated intestinal loops, and ultrasound shows a fibrotic membrane in the anorectal canal. Within minutes of removing the fibrotic tissue, the patient passes large amounts of meconium, and the abdominal distention decreased. Which of the following best describes the patient's condition?

A. Anal stenosis
B. Rectal atresia
C. Rectovaginal fistula
D. Membranous atresia
E. Anorectal agenesis

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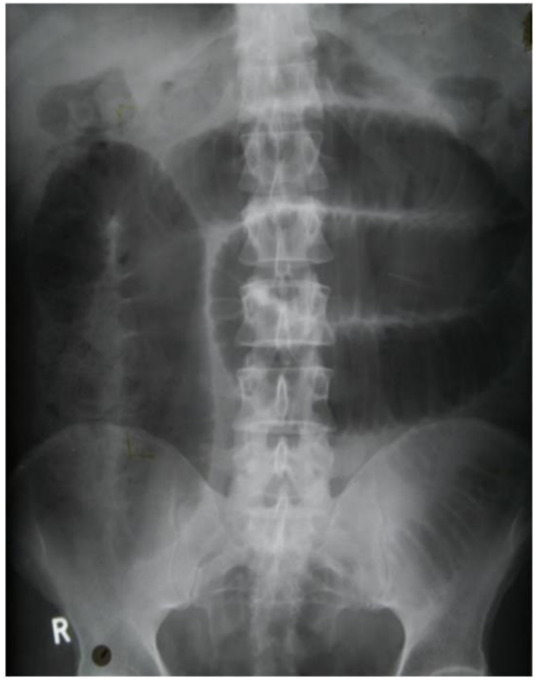
E. Anorectal agenesis

SOM.MK.1.BPM2.3.DM.1.ANAT.1109

Describe the developmental mechanism of imperforate anus, anal agenesis, anal stenosis, membranous atresia of the anus, rectal agenesis, and anorectal agenesis

A 37-year-old woman comes to the emergency department because of a 4-hour history of dull, worsening abdominal pain associated with nausea and vomiting. She has a history of abdominal surgeries. Physical exam shows a soft, distended abdomen with diffused tenderness. Imaging is done for suspected bowel obstruction (see attached). Diagnosis is confirmed and treatment is started. Visceral afferent fibers carrying pain from the affected bowel will most likely travel back to the spinal cord following which of the following neurons?

A. Somatic afferent neurons
B. Visceral afferent neurons
C. Sympathetic efferent neurons
D. Parasympathetic efferent neurons
E. Somatic efferent neurons



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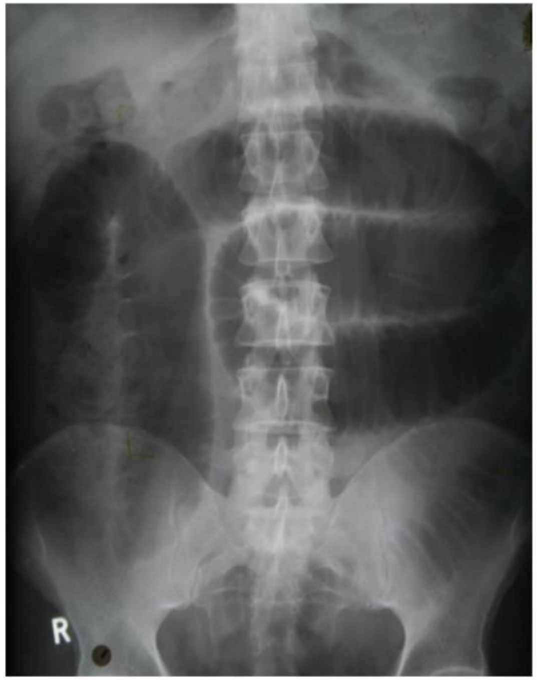

SOM.MK.1.BPM2.3.DM.1.ANAT.1113 Describe the sympathetic, parasympathetic and sensory innervation of the foregut, midgut and hind gut.

A 26-year-old man comes to the physician because of 4 hours of episodic abdominal pain. The pain typically occurs a few hours after eating and often wakes him up at night. It is relieved by drinking a glass of water and eating something bland. He has no history of nausea, vomiting or diarrhea but has recently noticed that some of his stools have been “almost black” in color. An upper GI endoscopy shows the presence of a duodenal ulcer. Posterior spread of this ulcer through the duodenal wall can cause erosion of the artery that lies directly behind this part of the duodenum. Which of the following structures is most likely to be affected because of this erosion?

A. Liver, stomach, duodenum and pancreas
B. Liver, duodenum and pancreas
C. Spleen, duodenum and pancreas
D. Liver, spleen, stomach, duodenum and pancreas
E. Stomach, duodenum and pancreas

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D. Liver, spleen, stomach, duodenum and pancreas

E. Stomach, duodenum and pancreas

SOM.MK.1.BPM2.3.DM.1.ANAT.1138 Describe the blood supply and the venous and lymphatic drainage of the duodenum, jejunum and ileum.

A 35-year-old man comes to the emergency department because of a 6-hour history of severe abdominal pain localized to the right lower quadrant. He has also experienced nausea and a single episode of vomiting. He appears uncomfortable and intermittently curls up in pain. His temperature is 37.8°C (100.04°F). Abdominal examination shows a "sausage-shaped" mass palpable in the right quadrant along with hyperactive bowel sounds and absence of rebound tenderness. Which of the following is the most likely diagnosis?

A. Gastric outlet obstruction
B. Acute pancreatitis
C. Intussusception
D. Diverticulitis
E. Volvulus

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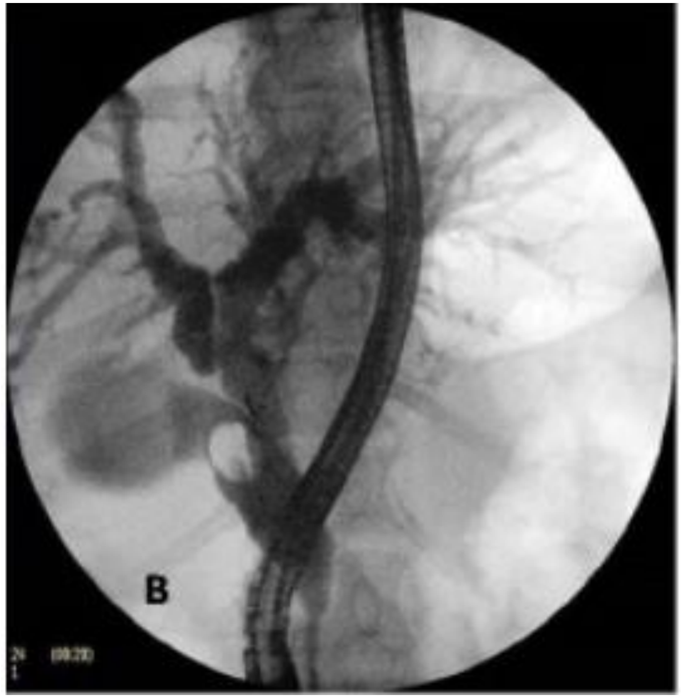
B. **Acute pancreatitis**
C. **Intussusception** ✓
D. **Diverticulitis**
E. **Volvulus**

SOM.MK.1.BPM2.3.DM.1.ANAT.1147
Descibe intussusception and its clinical presentation

A 54-year-old woman comes to the emergency department because of a 1-month history of right upper quadrant abdominal pain associated with yellowing of the eyes and intermittent fevers that started last week. She also reports dark colored urine, pale stool and jaundice for the past 2 days. Physical exam shows scleral jaundice and tenderness in the right upper quadrant. Laboratory studies show elevated bilirubin and liver enzymes. Ultrasound shows slight dilation of the intrahepatic ducts, hepatic bile ducts and gallbladder. ERCP shows a stone in the cystic duct, with no other stones along the biliary tree. Which of the following structures of the biliary tree is most likely extrinsically compressed by the stone?

A. Cystic duct
B. Gallbladder
C. Ampulla of Vater
D. Common bile duct
E. Common hepatic duct



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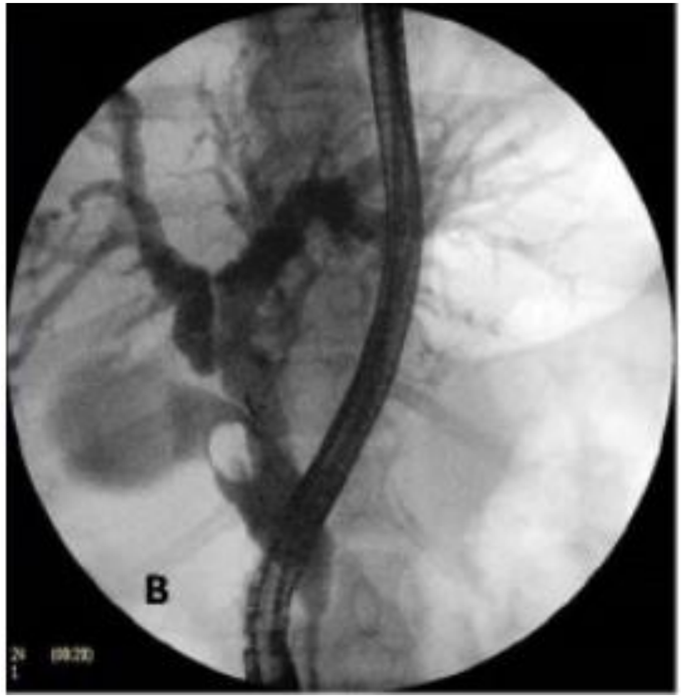

SOM.MK.1.BPM2.3.DM.1.ANAT.1180
Explain the pathway taken by an endoscope from mouth to the biliary tree during Endoscopic Retrograde Cholangio Pancreatography (ERCP) to remove a gallstone

A 74-year-old woman comes to the physician because of a 3-day history of epigastric pain. She reports having a chest pain that radiated to the back for 4 weeks prior to the abdominal pain. She has a history of hypertension, hyperlipidemia and GERD. Physical exam shows mild epigastric tenderness. Upper GI endoscopy shows large ulcerated tumor in the body of the stomach. Pathology exam confirms diagnosis of poorly differentiated adenocarcinoma. She undergoes curative gastrojejunostomy with removal of the distal end of the stomach. During surgery, the surgeon exposes the gastroesophageal junction to begin dissecting through the nearby omentum. Which of the following ligaments is most likely be initially dissected first to gain entrance into the lesser sac?

A. Falciform
B. Gastrocolic
C. Gastrosplenic
D. Hepatogastric
E. Hepatoduodenal

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B. **Gastrocolic**
C. **Gastrosplenic**
D. **Hepatogastric** ✔
E. **Hepatoduodenal**

SOM.MK.1.BPM2.3.DM.1.ANAT.1125
Describe the following ligaments, their embryonic origin, peritoneal coverings and their function: falciform, round ligament of the liver (ligamentum teres hepatis), coronary, triangular, median, medial and lateral umbilical, greater omentum, gastrocolic, gastrosplenic, splenorenal ligaments, lesser omentum (hepatoduodenal and hepatogastric ligaments) and suspensory ligament of the duodenum (Treitz).

A 58-year-old woman comes to the physician because of a 2-week history of epigastric pain and postprandial fullness. The pain extends along the left costal margin to the left shoulder and occurs after meals. Her symptoms started 2 years ago but have progressively worsened in intensity and frequency over the past 6 months. Her blood pressure is 130/86mmHg, pulse is 90/min, respirations are 15/min and temperature is 37.2oC (99oF). Physical exam shows mild tenderness in the epigastrium and left hypochondrium. An abdominal X ray shows gastric bubble posterior to the cardiac shadow. Barium study confirms normal location of distal esophagus. Which of the following best explains the patient's presentation?

A. Defect in upper esophageal sphincter
B. Aneurysm in the arch of the aorta
C. Enlargement of left main bronchus
D. Defect in the lower esophageal sphincter
E. Enlargement of the diaphragmatic hiatus

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D. Defect in the lower esophageal sphincter
E. Enlargement of the diaphragmatic hiatus

SOM.MK.1.BPM2.3.DM.1.ANAT.1129
Describe sliding and paraesophageal hiatal hernia

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Spring 2026

Biochemistry 9 Questions

Glycolysis, Gluconeogenesis, PDH and TCA cycle, ETC, Glycogen metabolism

A 25-year-old man is brought into the emergency department after being rescued by a search team. He was reported lost while exploring a cave by himself for the past 3 days. Physical examination shows moderate dehydration. Laboratory studies show hemoglobin: 14 g/dL (13.5-17.5), glucose: 73 mg/dL (70-100), sodium: 156 mEq/L (136-146), potassium: 3.9 mEq/L (3.5-5) and chloride: 96 mEq/L (95-105). He says that he had his last meal the night before the cave exploration. Which of the following enzymes is active in maintaining the plasma glucose levels over the past 3 days?

A. Citrate synthase
B. Fructose 2,6-bisphophatase
C. Hexokinase
D. Phosphofructokinase-1
E. Glucokinase
F. Pyruvate kinase

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SOM. MK.III. BPM2.3. DM.2.BCHM. 1106: Outline the metabolic changes that occur during the feed/fast cycle.

SOM. MK.I.BPM2.2.DM.3.BCHM.1141: Describe the reaction catalyzed by glucose 6-phosphatase and outline the release of glucose into the blood. Discuss how the deficiency of this enzyme affects both gluconeogenesis and glycogen degradation.

A 15-year-old girl is brought to the physician by her parents for a follow-up examination. Her fasting serum glucose concentration is 5.8 mmol/L (Ref Range: 3.8 -5.6) on the past two occasions. A diagnosis of maturity-onset diabetes of the young (MODY-2) resulting from a deficiency of glucokinase (Hexokinase IV) is made. Which of the following best describes this enzyme compared to the other hexokinase isoenzymes?

A. Mainly found in liver and kidney
B. In the nucleus during gluconeogenesis
C. Inhibited by its product
D. Lower Vmax for glucose
E. Lower Km for glucose

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SOM. MK.I. BPM2.3. DM.2.BCHM.1094 Explain the role of glucokinase in regulating blood glucose levels. Predict the effect of mutation in pancreatic glucokinase (MODY-2) on blood glucose levels.

A. Mainly found in liver and kidney
B. In the nucleus during gluconeogenesis
C. Inhibited by its product
D. Lower Vmax for glucose
E. Lower Km for glucose

A 14-year-old girl is brought to the physician by her parents because of a 2-week history of muscle pain and fatigue. She has had leg cramps, intermittent pain and weakness during exercise and after jogging. Physical examination shows bilateral thigh and calf muscle stiffness, pain and weakness. The remainder of the examination is normal. Urinalysis shows myoglobin. Surface electromyogram and ischemic forearm muscle exercise test are abnormal. A muscle biopsy shows focal accumulation of glycogen with a normal branching pattern. Which of the following sets of laboratory findings most likely corresponds with the patient's diagnosis?

A. I
B. II
C. III
D. IV
E. V

	Hemoglobin in urine	**Serum Glucose**	**Serum Creatine Kinase**	**Serum Transaminases (ALT, AST)**
I	Absent	Low	Normal	Elevated
II	Absent	Normal	Elevated	Normal
III	Present	Normal	Normal	Normal
IV	Present	High	Low	Elevated
V	Absent	Low	Elevated	Normal

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II	Absent	Normal	Elevated	Normal
III	Present	Normal	Normal	Normal
IV	Present	High	Low	Elevated
V	Absent	Low	Elevated	Normal

SOM. MK.I.BPM2.3.DM.2.BCHM.1166 Explain basis for clinical manifestations of type V (McArdle) glycogen storage disease: decreased exercise tolerance and lack of lactate increase in blood after exercise.

A 55-year-old man comes to the physician for a routine annual medical examination. Laboratory studies show normal liver and renal function tests. However, his lipid profile shows total cholesterol concentration ~~is~~ of 240 mg/dL (<200) , LDL-cholesterol concentration of 200mg/dl (<160) and HDL-cholesterol concentration of 25mg/dl (40-60). He is prescribed a statin drug to improve his lipid profile. This drug inhibits the synthesis of mevalonate required for effective production of cholesterol resulting in a decreased synthesis of a component associated with one of the complexes of the Electron Transport Chain. Which of the following steps in the Electron Transport Chain would most likely be directly affected in this patient?

A. Complex-I to complex-III
B. Complex-III to complex-IV
C. Complex-IV to oxygen
D. Cytochrome c to complex-IV
E. Fo-F1 ATP synthase

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SOM. MK.I.BPM2.2.DM.3.BCHM.1125 Differentiate between: direct inhibitors of the ETC, uncouplers of oxidative phosphorylation, ATP synthase inhibitors (oligomycin) and inhibitors of the ATP/ADP translocase

A 45-year-old man comes to the emergency department because of a 1-day history of worsening confusion and difficulty walking. He has a history of multiple admissions for alcohol induced seizures and withdrawal episodes over the past year. Physical examination shows confusion, ataxia and nystagmus. He is prescribed a vitamin supplementation before dextrose infusion. This vitamin is most likely a coenzyme for which of the following enzyme catalyzed reactions?

A. Citrate to isocitrate
B. Pyruvate to lactate
C. Pyruvate to oxaloacetate
D. Alpha-ketoglutarate to succinyl CoA
E. Glucose 6-phosphate to 6-phosphogluconate
F. Glycogen to glucose 1-phosphate

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SOM. MK.I.BPM2.2.DM.3.BCHM.1112 Discuss consequences of thiamine deficiency and its effect on PDH and αKGDH complexes in Wernicke-Korsakoff syndrome and beriberi.

An 18-month-old boy is brought to the physician because of a 6-month history of failure to thrive and tires quickly when playing. He is at 50th percentile for height and weight. Physical examination shows a protuberant abdomen with a palpable liver margin 2-cm below the right costal margin. Laboratory studies show hyperuricemia, lactic acidemia, elevated serum triacylglycerol (TAG) and hypoglycemia. Which of the following enzyme deficiency disorders is most likely in this patient?

A. Liver glycogen phosphorylase
B. Phosphofructokinase-1
C. Glycogen branching enzyme
D. Glycogen debranching enzyme
E. Glucose 6-phosphatase
F. Glucokinase

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C. Glycogen branching enzyme
D. Glycogen debranching enzyme
E. Glucose 6-phosphatase ✔
F. Glucokinase

SOM. MK.I.BPM2.3.DM.2.BCHM.1163 Describe type I (von Gierke) glycogen storage disease including the metabolic basis for severe hypoglycemia, hepatomegaly and lactic acidosis

A 51-year-old woman comes to the physician for a follow-up examination. She has a 6-month history of type-2 diabetes mellitus and is on metformin. She says she has episodes of tachycardia, dizziness and sweating which improves with food intake. Physical examination shows no abnormalities. Laboratory studies show a fasting blood glucose of 54 mg/dL (70-100). Which of the following sets of metabolic changes is most likely observed in the liver during her symptomatic periods?

A. I

B. II

C. III

D. IV

E. V

	PFK-1	PFK-2	Fructose bisphosphatase-2	Fructose 2,6-bisphosphate levels	Fructose 1,6-phosphatase
I	+	+	-	High	-
II	-	-	+	Low	+
III	-	+	+	Low	+
IV	-	-	+	High	+
V	+	-	+	Low	-

A 51-year-old woman comes to the physician for a follow-up examination. She has a 6-month history of type-2 diabetes mellitus and is on metformin. She says she has episodes of tachycardia, dizziness and sweating which improves with food intake. Physical examination shows no abnormalities. Laboratory studies show a fasting blood glucose of 54 mg/dL (70-100). Which of the following sets of metabolic changes is most likely observed in the liver during her symptomatic periods?

A. I
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	PFK-1	PFK-2	Fructose bisphosphatase-2	Fructose 2,6-bisphosphate levels	Fructose 1,6-phosphatase
I	+	+	-	High	-
II	-	-	+	Low	+
III	-	+	+	Low	+
IV	-	-	+	High	+
V	+	-	+	Low	-

SOM. MK.I. BPM2.3. DM.2.BCHM.1092; Appraise the role of AMP, ATP and fructose 2,6 bisphosphate on glycolysis.

A 12-year-old boy is brought to the physician because of a 2-month history of painful cramping and weakness during his exercise classes at school. He also says his urine is dark-colored. Laboratory studies show increased serum creatine kinase CK–MM. Result from a forearm ischemic test shows absence of lactate increase during exercise. A diagnosis of an inherited deficiency of a specific enzyme is made. Which of the following best describes the regulation of this enzyme?

A. Inhibited by calcium
B. Activated by AMP
C. Inhibited by glycogen phosphorylase kinase
D. Activated by ATP
E. Inhibited by phosphorylation

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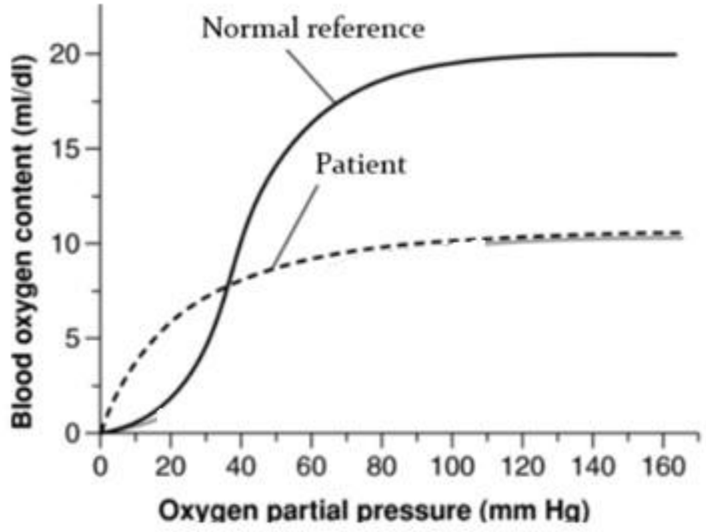
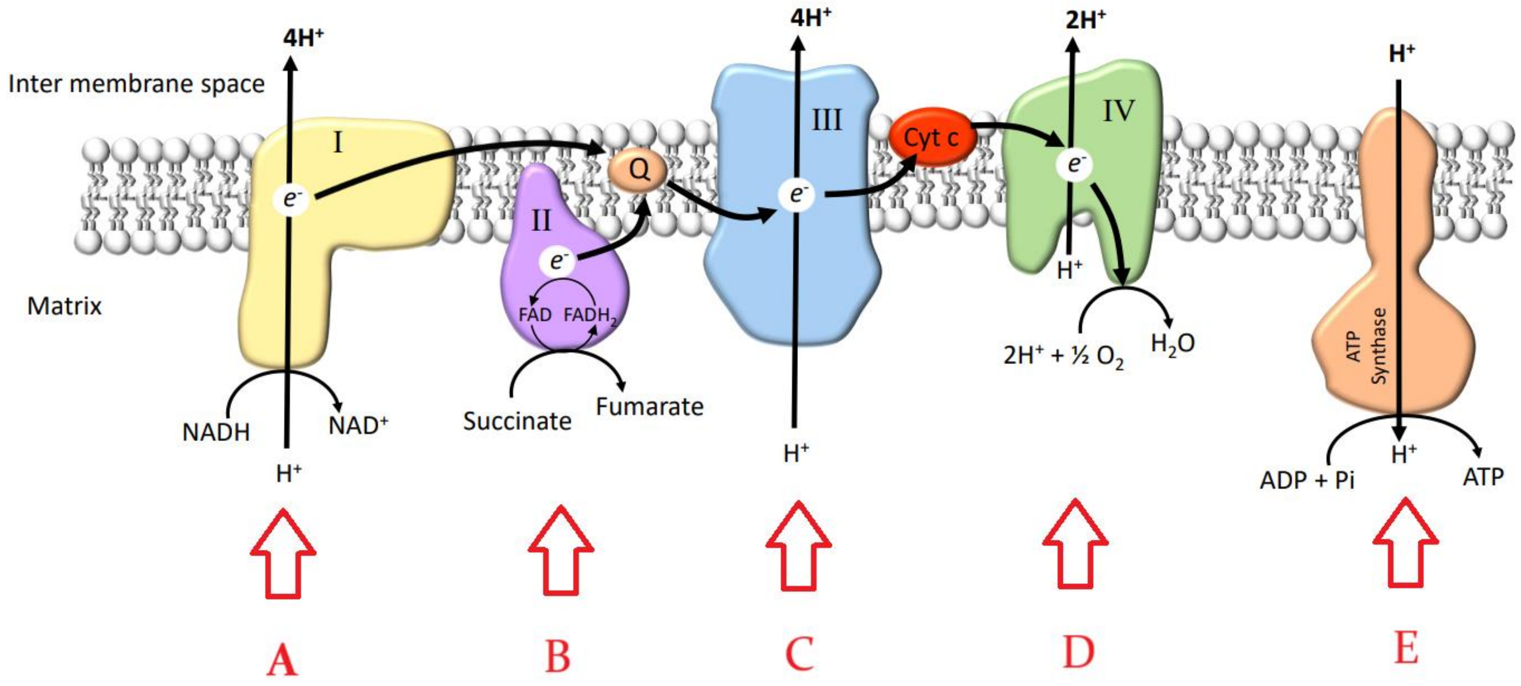
SOM. MK.I.BPM2.3.DM.2.BCHM.1158 Compare and contrast glycogen degradation in liver and muscle with reference to: hormones, metabolic state, and fate of glucose 6-P.

A. Inhibited by calcium
B. Activated by AMP
C. Inhibited by glycogen phosphorylase kinase
D. Activated by ATP
E. Inhibited by phosphorylation

A 35-year-old man is brought to the emergency department after being rescued from a burning house. He experiences a severe headache before losing consciousness on the way to the hospital. The expected changes in the hemoglobin-oxygen dissociation curve in this patient are shown in the graph. Which of the following components of the electron transport chain shown in the image is most likely inhibited in this patient?



A. Complex labeled A
B. Complex labeled B
C. Complex labeled C
D. Complex labeled D
E. Complex labeled E



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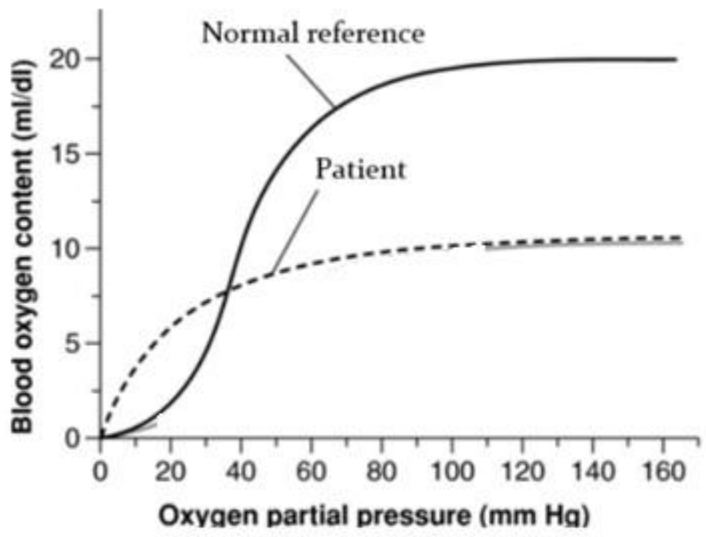
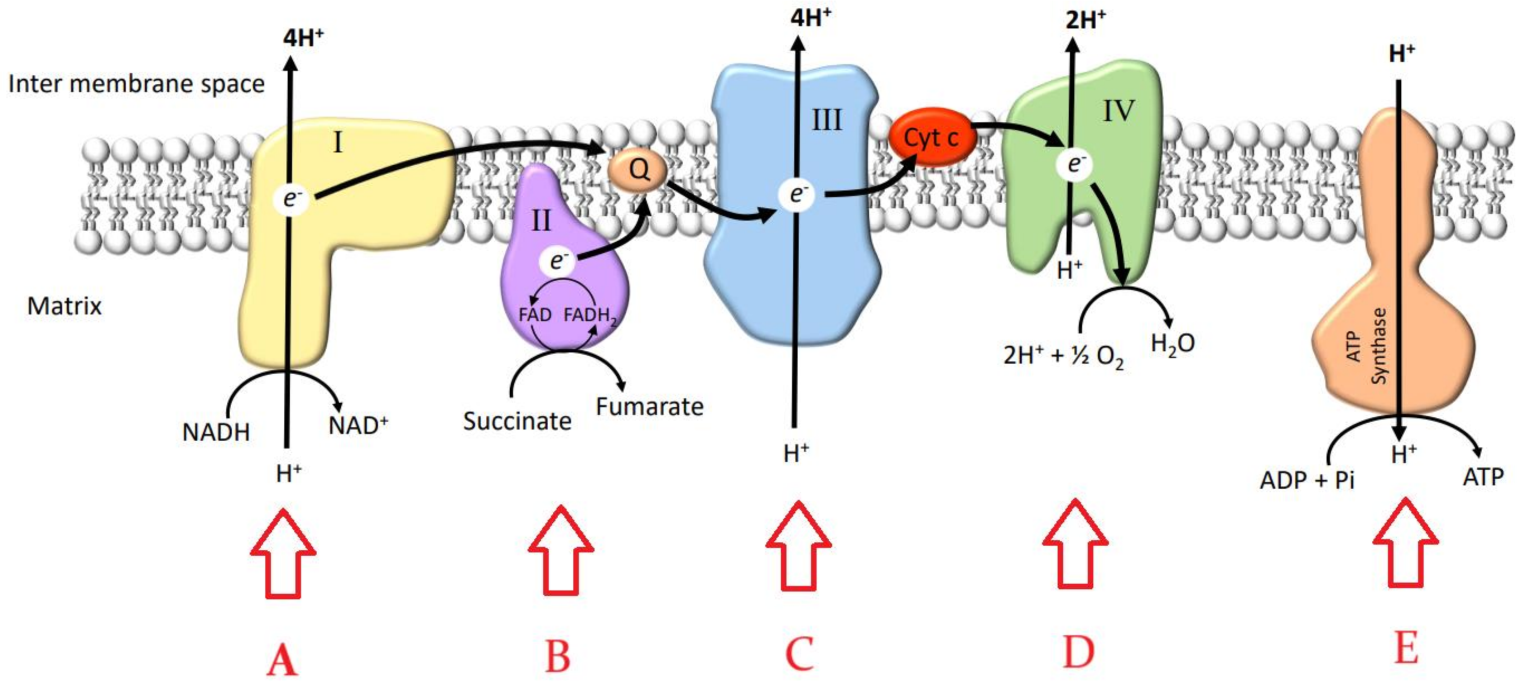


A. Complex labeled A
B. Complex labeled B
C. Complex labeled C
✓ D. Complex labeled D
E. Complex labeled E



SOM. MK.I.BPM2.2.DM.3.BCHM.1124; Discuss the mode of action/effects of specific inhibitors of the ETC (including antimycin A, azide, hydrogen sulphide, cyanide, rotenone, piericidin A, carbon monoxide)